# DGCA AVIATION METEOROLOGY

## MASTER FORMULA & EXAM CHEAT SHEET — FOR DAIVAM MEHTA

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## 1. ICAO International Standard Atmosphere (ISA) Constants

| Parameter | MSL Standard Value | Variation / Rule of Thumb |
|---|---|---|
| **MSL Pressure** | 1013.25 hPa / 29.92 inHg / 760 mmHg | 1 hPa = 27 to 30 ft (Low levels: 30 ft/hPa) |
| **MSL Temperature** | +15.0°C (288.15 K) | Decreases 1.98°C (~2°C) per 1000 ft (6.5°C/km) |
| **MSL Air Density** | 1.225 kg/m³ | Decreases with altitude, temp increase, moisture |
| **Tropopause Height (ISA)** | 36,090 ft (11,000 m) | Poles: 8-10 km (-45°C) \| Equator: 16-18 km (-80°C) |
| **Stratosphere Temp (ISA)** | -56.5°C (Isothermal up to 20 km) | Above 20 km temp increases (Ozone absorption) |
| **Speed of Sound (MSL)** | 661.47 kts (340.3 m/s) | LSS = 38.94 * sqrt(Temperature in Kelvin) |

## 2. Altimetry Formulas & Temperature Error Corrections

- **ISA Deviation: ISA Dev = OAT - (15 - 2 * (Alt / 1000))**
- **Altimeter Temp Error Correction: True Alt = Indicated Alt + (4 * (Alt/1000) * ISA_Dev)**
- **High to Low or Warm to Cold Rule: True Altitude is LOWER than Indicated. 'High to Low, Look out below!'**
- **QNH to QFE Conversion: QNH = QFE + (Field Elevation / 30 ft)**
- **Pressure Altitude: PA = Elevation + (1013.25 - QNH) * 30**
- **Density Altitude: DA = Pressure Altitude + [120 * (OAT - ISA_Temp_at_PA)]**

## 3. Lapse Rates & Cloud Base Determination

- **Dry Adiabatic Lapse Rate (DALR): 3.0°C / 1000 ft (9.8°C / km or 1°C / 100 m) — Constant**
- **Saturated Adiabatic Lapse Rate (SALR): ~1.5°C / 1000 ft (4°C to 9°C / km) — Varies with Temp & Moisture**
- **Dew Point Lapse Rate in Unsaturated Air: 0.5°C / 1000 ft**
- **Convective Cloud Base (AGL ft): Base = [(T - Td) / 2.5] * 1000 = 400 * (T - Td)**

## 4. Wind & Circulation Dynamics

- **Geostrophic Wind Formula: Vg = 1 / (2 * omega * rho * sin(phi)) * (dP / dn)**
- **Crosswind Component: Crosswind = Wind_Speed * sin(Angle between Runway and Wind)**
- **Headwind Component: Headwind = Wind_Speed * cos(Angle between Runway and Wind)**
- **Buys Ballot's Law: Northern Hemisphere — Stand with back to wind, LOW pressure is on your LEFT.**
- **Friction Layer Effect: Climbing out in NH -> Wind VEERS (clockwise) and INCREASES in speed.**

## 5. The 15 Golden DGCA Exam Mnemonics for Daivam Mehta

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| **1. Altimeter Pressure Rule: HIGH to LOW, look out BELOW. (Altimeter OVERREADS; true altitude is LOWER than indicated).** |
| **2. Altimeter Temp Rule: COLD air COMPRESSES, WARM air EXPANDS. (Flying cold -> aircraft is lower than indicated).** |
| **3. Buys Ballot's Law (NH): Back to the wind, LOW is on the LEFT (Northern Hemisphere). South: LOW is on the RIGHT.** |
| **4. Wind Direction Shift: VEER = Clockwise (Right) \| BACK = Counter-Clockwise (Left).** |
| **5. Boundary Layer Friction: Climb out of friction: Wind VEERS and SPEEDS UP. Descending into friction: Wind BACKS and SLOWS.** |
| **6. Cloud Base Rapid Calc: 400 * Spread = Base in ft AGL. (Spread = Surface Temp - Dew Point).** |
| **7. Density Altitude Thumb: DA = PA + (120 * ISA Deviation). High, Hot & Humid = Poor Performance!** |
| **8. Icing Sweet Spots: Clear Ice: 0°C to -10°C (Large droplets / Cumuliform). Rime Ice: -10°C to -20°C (Small droplets / Stratiform).** |
| **9. Thunderstorm Stages: C-M-D: Cumulus (Updrafts only) -> Mature (Updrafts + Downdrafts + Rain) -> Dissipating (Downdrafts only).** |
| **10. Frontal Sequence (Warm): Cirrus -> Cirrostratus (Halo) -> Altostratus -> Nimbostratus (Continuous rain) -> Stratus/Fog.** |
| **11. Cyclone Wind Rotation: Northern Hemisphere: COUNTER-CLOCKWISE & INWARD. Southern: CLOCKWISE & INWARD.** |
| **12. Fog Formation Magic: Radiation Fog needs: High RH, Clear skies, Calm to Light wind (2-8 kts), Land only (NEVER over sea).** |
| **13. Jet Stream Axis Rule: Cold air to the LEFT in Northern Hemisphere -> Maximum CAT is on the COLD (polar) and LOW side of jet core.** |
| **14. CAVOK Minimums: Visibility >= 10 km, NO cloud below 5,000 ft or MSA (whichever higher), NO CB / TCU, NO significant weather.** |
| **15. Mountain Wave Signs: Lenticular (lens) clouds at crests, Roll/Rotor clouds in turbulent lower zone, Cap cloud on peak.** |